Click, Clack, Clunk: Development and Evaluation of a Low-Cost Developmental Dysplasia of the Hip (DDH) Simulation Task Trainer

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Background

DDH:

- Rare developmental joint abnormality
- Increases lifetime risk of osteoarthritis and arthroplasty [1,2]

Early detection via Barlow/Ortolani maneuvers enables proactive management[3]

Problem:

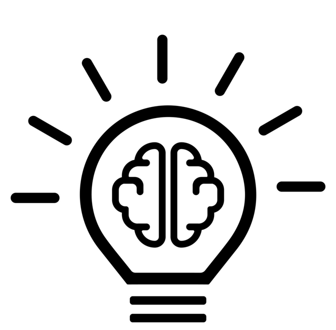
- Low incidence limits hands-on provider experience → **poor exam sensitivity in practice** (~36%) [4]
- **Task trainers can build experience**, but **current options are expensive** ($2,110+), limiting access [1]

Goal:

Develop a low-cost task trainer to enhance Barlow/Ortolani proficiency

Methods

Part 1: Engineering & Design Process



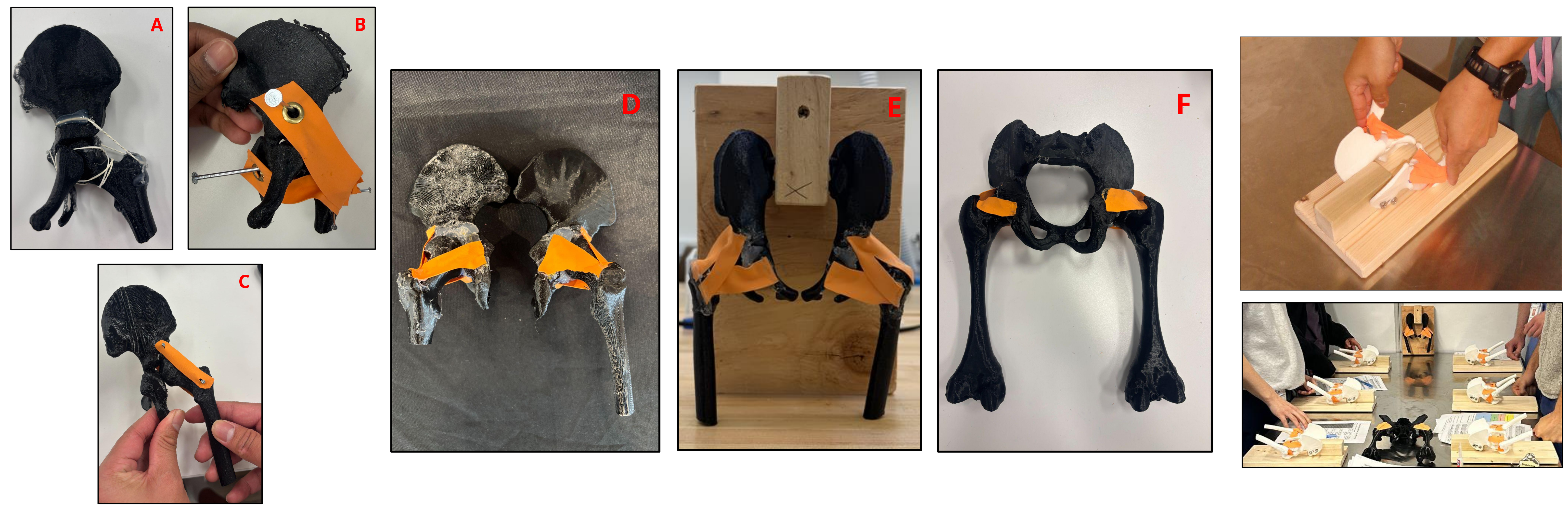
Physiological mimicry
Anatomical mimicry
Ease of Assembly
Cost
Component Count

Part 2: Evaluation Process

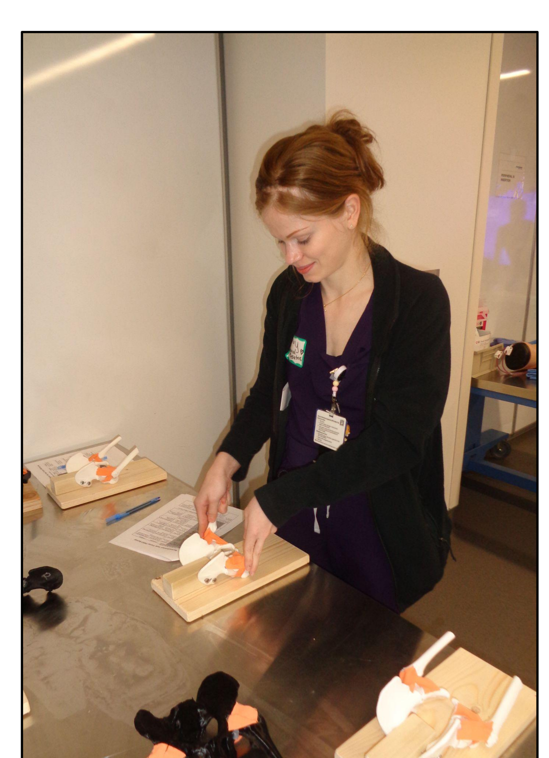
Pre-Survey → Training → Post-Survey

- **Pre**: Baseline self-evaluation on competency[5,6]
- **Train**: Simulation OSCE with deliberate practice + real-time feedback[7,8,9]
- **Post**: Repeat self-evaluation and scored via checklist[5,10,11]

Development of Task Trainer



Results



Domain	Pre-session % ≥ 2	Post-session % ≥ 2	b (gained)	c (lost)	p-value	
Anatomy identification†	84.4%	100.0%	5	0	0.062	ns
Indications & contraindications	62.5%	100.0%	12	0	<0.001	***
Maneuver purpose†	87.5%	100.0%	4	0	0.125	ns
Interpretation of findings	68.8%	100.0%	10	0	0.002	**
Positioning & technique	68.8%	100.0%	10	0	0.002	**
Troubleshooting	28.1%	96.9%	23	1	<0.001	***

Testing done at UTSW Pediatric Intern Boot Camp

- 32 residents
- McNemar's exact test, pre- vs. post-training competency (≥2 rating)
- **Largest gains: procedural domains**
  - Positioning/Technique (68.8%→100%, p<0.01)
  - Troubleshooting (28.1%→96.9%, p<0.001)
- **Smallest gains: cognitive domains**
  - Anatomy ID and Maneuver Purpose showed ceiling effects (already ≥84% competent at baseline)

Discussion

- Providers improved most in executing the maneuver
- Anatomy/indications were already well-taught pre-residency
- Highlights a procedural, not knowledge, gap
- Supports efficacy of both trainer design and instructional program

Future Directions:

- Expert review of design fidelity
- Pre/post clinical performance study
- Continued task trainer design iteration

References